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UNIVERSITY OF SRI JAYEWARDENEPURA - FACULTY OF APPLIED SCIENCES
B.Sc. General Degree Second Year First Semester Course Unit Examination – Sept/Oct 2023

DEPARTMENT OF MATHEMATICS

MAT 211 1.0 Linear Algebra II

Time: One hour. No. of questions: 04 No. of pages: 05 Total marks: 80

Instructions:

- Answer all questions in the given space.
- You may bring **one A4 Sheet** of information for the exam.
- Basic scientific calculators are allowed.
- Usual notation applies throughout.

1. Let $A = \begin{pmatrix} 2 & 5 \\ 5 & 2 \end{pmatrix}$.

(i) Show that $v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $v_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ are eigenvectors of A . [4]

(ii) (a) For any two vectors in $u, v \in \mathbb{R}^2$, taking the norm $\langle u, v \rangle = u \cdot v$ find an orthogonal matrix P which diagonalises A .

(b) Give the diagonal matrix D from this diagonalisation. [15]

(iii) Use the diagonalisation in (ii) to find A^4 .

[8]

2. (i) Write the quadratic form $f(x, y) = 7x^2 + 12xy - 2y^2$ in new variables x' and y' so that no cross-product term xy is present. [15]

- (ii) Hence identify the conic section of $7x^2 + 12xy - 2y^2 = 5$. [10]

3. (i) Given that the eigenvalues of the matrix A are $\lambda_1, \dots, \lambda_n$, find the eigenvalues of $-A$.

[6]

- (ii) Show that similar matrices do have same eigenvalues.

[

4. What are the possible Jordan canonical forms for the following characteristic polynomials:

[16]

(i) $(\lambda - 2)^3(\lambda + 5)$

(ii) $(\lambda - 6)^4$.

